



01000101 01001110 01000011 01001111 01000100 01001001 01001110 01000111 00100000 01010100 01000101 010

ROOT@SEC-TERM:~# **ENCODING BASICS & TECHNICAL REFERENCE**

[OK] Initialized data conversion specs, standard protocols, and detection signatures.

01 //Hashing vs. Encoding vs. Encryption

Property	Encoding	Hashing	Encryption
Primary Purpose	Data compatibility / usability across systems [cite: x]	Data integrity & one-way verification [cite: x]	Data confidentiality / privacy [cite: x]
Reversibility	Two-way (No key) [cite: x]	One-way (Non-reversible) [cite: x]	Two-way (Requires key) [cite: x]
Secret Key Required?	No [cite: x]	No [cite: x]	Yes [cite: x]
Output Size	Variable (proportional to input) [cite: x]	Fixed size (algorithm defined) [cite: x]	Variable (proportional to input) [cite: x]
Security Level	Zero Security [cite: x]	Cryptographic Integrity [cite: x]	Cryptographic Confidentiality [cite: x]

02 //Character Encodings (Text Representation)

Character encodings map binary bits to human-readable characters across different languages and systems [cite: x].

- **ASCII**: 7-bit standard, 128 English chars & symbols [cite: x].
 - **ISO-8859-1 (Latin-1)**: 8-bit extension for Western Europe [cite: x].
 - **Windows-1252**: Legacy 8-bit standard extending Latin-1 [cite: x].
- Unicode Standards:** [cite: x]

 - **UTF-8**: Var-width (1-4B). ASCII compatible. Web standard [cite: x].
 - **UTF-16**: Var-width (2/4B). Standard in WinOS & Java/JS [cite: x].
 - **UTF-32**: Fixed-width (4B). High memory overhead [cite: x].

03 //Data Transfer & Binary Encodings

Used to represent binary or non-ASCII data in safe, printable ASCII text formats for network transmission (HTTP, email, URLs) [cite: x].

- Base64 Mechanics** [cite: x]

 - **Alphabet**: A-Z, a-z, 0-9, +, / (64 chars) [cite: x]
 - **Padding**: = (When input bytes % 3 != 0) [cite: x]
 - **Ratio**: ~+33% overhead (3B → 4 chars) [cite: x]
- Base32 Mechanics** [cite: x]

 - **Alphabet**: A-Z, 2-7 (32 chars, Case-insensitive) [cite: x]
 - **Padding**: = [cite: x]
 - **Use Case**: Case-insensitive/legacy systems [cite: x]

04 //Command-Line Reference & Identification

Base64 Operations [cite: x]

```
# Encode string without newline
echo -n "TryHackMe" | base64 [cite: x]

# Decode Base64 string
echo "VHJ5SGFja011" | base64 -d [cite: x]

# Encode / Decode Files
base64 input.txt > encoded.txt [cite: x]
base64 -d encoded.txt > decoded.txt [cite: x]
```

Base32 Operations [cite: x]

```
# Encode string
echo -n "TryHackMe" | base32 [cite: x]

# Decode Base32 string
echo "OVRXSzRZE1I=" | base32 -d [cite: x]
```

URL / Percent Encoding [cite: x]

Replaces non-ASCII/reserved chars with %XX (e.g. Space → %20, ' → %27) [cite: x]

```
# Encode via Python
python3 -c "import urllib.parse; print(urllib.parse.quote('admin' OR 1=1--'))" [cite: x]

# Decode via Python
python3 -c "import urllib.parse; print(urllib.parse.unquote('admin%27%20R%201%3D1--'))" [cite: x]
```

Hexadecimal Encoding (Base16 / Hex Dump) [cite: x]

Converts raw bytes to 2 hex digits (0-9, a-f) [cite: x]

```
# String to Hex & Hex Back to Plaintext
echo -n "TryHackMe" | xxd -p [cite: x]
echo "5472794861636b4d65" | xxd -r -p [cite: x]
```

05 //Identifying Encodings in Penetration Testing

Encoding	Key Identification Signatures	Example Pattern
Base64 Trait [cite: x]	Length is multiple of 4. Ends with = or == padding. Mixed-case alphanumeric, +, and / [cite: x].	VHJ5SGFja011==
Base32 Trait [cite: x]	Uppercase letters A-Z and digits 2-7 only. Ends with up to six = padding characters [cite: x].	OVRXSzRZE1I=
Hex / Base16 Trait [cite: x]	Length is an even number. Contains only characters 0-9 and A-F (or a-f) [cite: x].	5472794861636b4d65
URL Encoding Trait [cite: x]	Frequently contains % followed by two hex characters [cite: x].	%3Cscript%3E [cite: x]